

## LESSON 9-4 • SESSION 2 • PRACTICE • FORM A

**Working Backward to a Goal**

Learning target – I can work backward from a goal to find the input an equation needs. • 6.EE.C.9

**REMEMBER**

- **Work backward:** put the goal in the equation, then undo the math.
- With a fee, **subtract** it first, then **divide**:  $20 = 4n + 8$ , so  $20 - 8 = 12$  and  $12 \div 4 = 3$ .
- If only whole items make sense, check whether you must **round up**.

**WORKED EXAMPLE**

Tickets cost \$6. A club needs at least \$100. How many tickets?

**Step 1** Put in the goal:  $100 = 6n$ .

**Step 2** Divide:  $100 \div 6 \approx 16.7$ .

**Step 3**  $16 \times 6 = \$96$  is too low;  $17 \times 6 = \$102$  is enough.

**Answer 17 tickets**

**WORDS** **work backward** (*trabajar hacia atrás*) • **inverse operation** (*operación inversa*) • **goal** (*meta*)

**PRACTICE**

show your work • label units

1. Solve for  $x$ .

a.  $9x = 126$  \_\_\_\_\_

b.  $6x = 90$  \_\_\_\_\_

c.  $4x + 8 = 52$  \_\_\_\_\_

d.  $5x + 30 = 155$  \_\_\_\_\_

2. **School Garden.** A class sells painted flower pots for \$12 each. It needs to raise \$660.

a. Write an equation for the money  $m$  from  $p$  pots. \_\_\_\_\_

b. Put in the goal and work backward. \_\_\_\_\_ pots

c. If the class sells only 44 pots, what price must it charge? \_\_\_\_\_ dollars

3. **Read-a-thon.** Sponsors pay Leah \$3 for each page she reads. How many pages must she read to raise \$210?

**Hint** Divide the goal by the pay for one page.

Show your work.

**Answer** \_\_\_\_\_ pages

4. **Talent Show.** Tickets cost \$7 each. How many tickets must the club sell to raise \$350?

**Hint** Undo the multiplication.

Show your work.

**Answer** \_\_\_\_\_ tickets

5. **Charging Station.** A charger adds 9 miles of range every minute. Write an equation. How many minutes to add 135 miles?

**Hint** The goal replaces  $r$ . Then divide.

Show your work.

**Answer** \_\_\_\_\_ minutes

6. **Concert Tickets.** Tickets cost \$25 each. A group has \$310 and says it can buy 13 tickets. Does the plan work? What is the most it can buy?

Show your work.

Most tickets \_\_\_\_\_

7. **Class Trip Fund.** A class has \$90. It earns \$15 for each raffle ticket sold. It needs \$600 in all.

**Hint** Subtract what the class already has first.

- a. Write an equation for the money  $m$  after  $k$  tickets are sold. \_\_\_\_\_  
 b. How many tickets must be sold to reach \$600? \_\_\_\_\_

Show your work.

8. **Round Up.** A club earns \$8 for each ticket and needs \$300.

**Hint** You cannot sell part of a ticket.

- a. Solve  $8t = 300$ . \_\_\_\_\_  
 b. How many tickets must the club sell? \_\_\_\_\_

## FIND THE ERROR

explain the mistake, then fix it

9. **Renata's Work.** Wristbands cost \$4 each. Renata needs \$96. She wrote  $4b = 96$  and then answered 384 wristbands. Explain her mistake and find the correct number.

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## TEST PRACTICE

choose the best answer

10. Solve  $9n = 117$ .

**A** 108                      **B** 13                      **C** 126                      **D** 1,053

11. For  $c = 4n + 8$ , if  $c = 40$ , then  $n$  is –

**A** 12                      **B** 10                      **C** 8                      **D** 168

## SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|                       |                       |                       |                       |                       |                        |                       |                       |                       |                       |
|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <b>15</b><br><b>E</b> | <b>12</b><br><b>R</b> | <b>34</b><br><b>S</b> | <b>38</b><br><b>E</b> | <b>50</b><br><b>V</b> | <b>630</b><br><b>X</b> | <b>40</b><br><b>Q</b> | <b>55</b><br><b>I</b> | <b>37</b><br><b>Z</b> | <b>70</b><br><b>N</b> |
| #2b                   |                       | #3                    | #4                    | #5                    | #6                     | #7b                   | #8b                   |                       |                       |

## CHECK YOURSELF

☐ I can do this on my own    ☐ I need a little help    ☐ I have a question about #\_\_\_\_\_